

1. Introduction

This report documents the Sri Lanka Dengue Surveillance Dashboard, an interactive Quarto dashboard built to explore weekly, district level dengue case data reported across Sri Lanka. The dashboard is organised into five pages; Overview, Seasonality & Trends, District & Province Analysis, Outbreak Analysis and Data Explorer. Each page brings together a set of interactive charts, maps and tables designed to help public health analysts understand the long term trend, seasonal pattern, geographic distribution and outbreak behaviour of dengue in Sri Lanka. This document explains the data source, the key performance indicators (KPIs), every chart on every page, how to read & interact with each visual and the epidemiological interpretation that can be drawn from it.

2. Data Source: the denguedatahub Package

The dashboard is built on top of the denguedatahub R package, developed and maintained by Dr. Thiyanga S. Talagala (Department of Statistics, University of Sri Jayewardenepura) with funding from the R Consortium's Infrastructure Steering Committee (ISC). The package's stated purpose is to provide the global research community with a unified, tidy format collection of dengue related data, gathered from multiple countries and merged with variables that support a better understanding of dengue transmission and reproducible dengue research.

For Sri Lanka specifically, the package includes web scraping utilities that retrieve the Weekly Epidemiological Reports published by the Epidemiology Unit, Ministry of Health, Sri Lanka and clean this information into a tidy data frame. The dataset used throughout this dashboard, `srilanka_weekly_data`, contains district wise weekly dengue case counts from late 2006 to 2025, with the columns `year`, `week`, `start.date`, `end.date`, `district`, and `cases`. In addition to the raw data, the package also ships data manipulation helpers (such as min-max normalisation functions) intended to support downstream visualisation and modelling work. The package is available on CRAN and GitHub and it also powers a companion Shiny application and documentation website, all part of the wider "Dengue Data Hub" initiative.

3. Dashboard Structure

The dashboard is arranged as five pages accessible from the top navigation bar ;

- Overview : Headline KPIs, the full national weekly time series, annual totals, the current year's seasonal position and a district map.
- Seasonality & Trends : Three plots that isolate the recurring within year seasonal pattern and compare years against each other.
- District & Province Analysis : Geographic breakdown of the case burden by district and by province, including two browsable heatmaps.
- Outbreak Analysis : Epidemic threshold and anomaly detection views used to identify unusual or outbreak level weeks.

- Data Explorer : A searchable, filterable, exportable data table of the underlying district level weekly records.

4. Key Performance Indicators (KPIs)

Four value boxes sit at the top of the Overview page. They give an at a glance summary of the scale, history and recency of the dengue burden before the person looks at any chart.

4.1 Total Cases (full data range)

This KPI shows the sum of all dengue cases recorded across every district and every year in the dataset (2006 to the most recent year - 2025). It represents the cumulative national dengue burden over the entire surveillance period. A very large number here communicates the sheer scale of dengue as a recurring public health problem in Sri Lanka, rather than an isolated event.

4.2 Peak Outbreak Year

This KPI identifies the single calendar year with the highest total number of reported cases nationally. It draws the viewer's attention immediately to the worst outbreak year on record (historically, this has been 2017, the year of Sri Lanka's largest dengue epidemic). This figure is a useful anchor for comparison: any other year's totals can mentally be benchmarked against this peak.

4.3 Cases in Latest Year

This KPI reports the total number of cases recorded in the most recent year present in the dataset. It tells the viewer how the most recent year compares to the historical peak and to the overall average, giving a quick read on whether the situation is currently severe, moderate or mild relative to history.

4.4 Districts Monitored

This KPI shows the count of distinct districts contributing data to the dashboard (26 districts across Sri Lanka's nine provinces). It confirms the geographic coverage of the surveillance data and reassures the viewer that the analysis is nationwide rather than partial.

5. Overview Section - Charts and Interpretation

The Overview page contains four visuals beneath the KPI row: a long run weekly time series, an annual bar chart, a seasonal comparison line chart and a district map.

5.1 Weekly Dengue Cases (2006–2025)

Description: This is the dashboard's master time series chart. It plots the national weekly case count as a red, shaded area/line running across the entire study period, with year boundaries marked by faint dotted vertical guides and year labels along the x axis. A toggle button lets the viewer switch between the raw "Weekly" series and a smoothed "4 week average" series. A range slider beneath the axis and quick zoom buttons (1 Year, 5 Years, All) allow the viewer to zoom into any period. A small annotation notes that the very first data point (week 52 of 2006) is an isolated single week, with continuous weekly data only beginning in 2007.

Interpretation: The chart shows that dengue transmission in Sri Lanka is not steady; it rises and falls in sharp seasonal waves nearly every year, with certain years, most visibly 2017 and 2019, producing case counts far above the typical range. Switching to the 4 week average view removes short term week to week noise and makes the underlying wave pattern and the relative height of each year's peak easier to compare. The long run view also shows that the general case level has shifted over the two decades, rather than staying flat, which is consistent with dengue becoming an increasingly endemic, rather than occasional disease in the country.

5.2 Annual Cases - Peak Year Highlighted

Description: A vertical bar chart showing total dengue cases for each calendar year. Every bar is coloured light tan except for the single peak year, which is highlighted in orange with a red outline and an arrow annotation marking it as the outbreak year.

Interpretation: This chart converts the noisy weekly series into one clean number per year, making it straightforward to rank years by severity. The highlighted bar immediately shows how much larger the worst year's total was compared to a typical year, often several times the case count of an average year underscoring how disproportionately damaging a single epidemic year can be relative to the rest of the record.

5.3 Seasonal Pattern (Avg by Week)

Description: A line chart comparing the latest year's weekly case counts (green line with markers) against the average of all other years at the same calendar week (grey dotted line with a light shaded area). The x axis is labelled by month rather than week number for readability.

Interpretation: This chart tells the viewer whether the most recent year is running above, below or in line with the typical seasonal pattern at each point in the calendar. Where the green line sits above the grey "average of other years" band, the latest year is experiencing an above normal dengue load for that time of year; where it sits below, transmission is currently lower than usual. Because the comparison excludes the current year's own data from the baseline, it avoids the current year distorting the benchmark it is being judged against.

5.4 Cases by District (Map)

Description: An interactive Leaflet map of Sri Lanka with a circle marker placed at each district's approximate centroid. Both the size and the colour of each circle scale with that district's total case count over the full data period (colour ranges from light blue for low burden districts through purple and red to dark red for the highest burden districts). Hovering or clicking a circle reveals the district name and its total case count and a legend in the bottom right corner explains the colour scale.

Interpretation: The map gives an immediate geographic read on where dengue is concentrated. Larger, darker red circles typically clustering around the Western Province (Colombo, Gampaha, Kalutara) indicate that dengue burden in Sri Lanka is heavily skewed toward the densely populated wet zone urban districts, rather than spread evenly across the island. Smaller, lighter circles in northern and eastern districts indicate a comparatively lower historical case count in those areas.

6. Seasonality & Trends Section - Charts and Interpretation

This page contains three plots that specifically isolate and examine the within year seasonal cycle of dengue transmission.

6.1 Weekly Dengue Seasonality - Average Cases & Historical Range

Description: For each epidemiological week (1-52), this chart plots the average weekly case count across all years (purple line with markers) together with a shaded ribbon showing the historical minimum to maximum range observed in that week across all years. Two translucent rectangles mark Sri Lanka's two monsoon periods, the South-West Monsoon (May-July) in orange and the North-East Monsoon (October-January) in blue and an annotation marks the single week with the highest average case count.

Interpretation: The purple average line traces a recurring within year seasonal cycle rather than a flat line, confirming that dengue transmission is seasonal and not random throughout the year. Because case counts tend to rise a few weeks after the monsoon rains (which create mosquito breeding sites), the peak in the average line is expected to fall shortly after one of the two shaded monsoon bands, most consistently following the South-West Monsoon. The width of the shaded min-max ribbon shows how much variation exists between years at the same calendar week, a wide ribbon around the peak week indicates that some years experience a much larger seasonal surge than others, which is exactly where major outbreak years like 2017 depart from the historical norm.

6.2 Year-on-Year Comparison - Top 5 vs. All-Years Range

Description: This chart overlays the weekly case trajectories of the five highest total years directly on top of one another (each drawn in a distinct colour), against a background teal ribbon showing the all years minimum to maximum range for context. Quick zoom buttons let the viewer focus on particular parts of the year.

Interpretation: Placing the five worst years side by side on the same calendar week axis reveals whether major outbreaks share a common seasonal shape (i.e., they typically peak in the same weeks as ordinary years, only at a much higher case level) or whether some big years peak earlier, later or have a fundamentally different shaped curve. When the top 5 lines rise well above the shaded "all-years range," it confirms that these years were genuine outliers, not simply years at the upper edge of the normal range.

6.3 Seasonal Subseries - Monthly Case Distribution Across All Years

Description: For each calendar month, this chart draws a box plot summarising the distribution of weekly case counts recorded in that month across every year in the dataset, together with a connecting dotted line of red diamond markers showing the mean weekly cases for that month.

Interpretation: This view is the clearest single confirmation of seasonality in the dashboard, because it separates "typical spread of values in a month" from "the month's average level." A pattern where the red mean line rises noticeably in certain months and falls in others demonstrates a real, recurring seasonal cycle rather than random month to month noise. Months with taller boxes and more outlier points above the box indicate months that are seasonally prone to sudden case spikes (i.e., higher outbreak risk) even though the median case count for that month may not be dramatically higher.

7. District & Province Analysis Section - Charts and Interpretation

This page contains four visuals: a ranked bar chart of districts, a province level trend line chart and two browsable heatmaps.

7.1 Top 10 Districts by Dengue Cases - Select Year or All Time

Description: A horizontal bar chart ranking the ten districts with the highest total case counts, from highest to lowest. Bars are shaded from dark red to pale pink according to case volume and each bar is labelled with its exact total. A dropdown menu above the chart lets the viewer switch the ranking between "All Years" (the full period total) and any single year in the dataset, with the bars and their order updating accordingly.

Interpretation: In the "All Years" view, the chart identifies which districts have consistently carried the largest share of the national dengue burden over the whole surveillance period typically the populous Western Province districts. Switching to individual years lets the viewer check whether the same districts dominate every year or whether the ranking shifts, which would indicate that outbreak years affect different parts of the country than the long run pattern suggests.

7.2 Province-Level Annual Trends - Linear & Log Scale

Description: A multi line chart showing each of Sri Lanka's nine provinces' total annual case counts over time, each province drawn in its own colour with a shared legend. Buttons above the chart toggle the y axis between a linear scale and a logarithmic scale.

Interpretation: On the linear scale, provinces with a much larger case burden (typically Western Province) visually dominate the chart, which can compress the visible detail of lower burden provinces near the bottom. Switching to the log scale rescales the y axis so that percentage changes are equally visible for both large and small provinces, making it possible to see whether a low burden province is also experiencing proportionally large year on year swings or trending up or down, even though its absolute case numbers stay well below Western Province's.

7.3 Weekly Dengue Intensity by Province - Browse by Year

Description: A heatmap with provinces on the y axis (ordered from lowest to highest total burden, bottom to top) and epidemiological week (labelled by month) on the x axis. Cell colour represents the number of cases reported in that province in that week, using a colour scale running from pale blue (low) through purple and orange to dark red (very high). A dropdown menu lets the viewer select which year's data is displayed.

Interpretation: Reading across a single row shows how that province's case intensity rises and falls through the selected year, while reading down a single column compares all provinces during the same week. Rows that turn deep red for extended stretches identify a province going through a sustained, high intensity transmission period; comparing this across different selected years shows whether outbreak weeks are geographically widespread (many rows turn red at once) or confined to just one or two provinces.

7.4 Weekly Dengue Intensity by District - Browse by Year

Description: The same style of browsable heatmap as 7.3, but broken down to the finer district level instead of province, with districts ordered from highest total historical burden (top) to lowest (bottom). A year selection dropdown again controls which year's weekly pattern is shown.

Interpretation: Because it operates at district resolution, this heatmap can reveal outbreak activity that is masked when data is aggregated to the province level, for example, a single high burden district driving a province's numbers while its neighbouring districts remain comparatively quiet. Comparing multiple years through the dropdown shows whether the same districts repeatedly light up at the same time of year (a stable seasonal pattern) or whether different districts become hotspots in different years (a more unpredictable, spatially shifting outbreak pattern).

Important note on using the two heatmaps: after zooming, hovering or selecting a different year in either the "Weekly Dengue Intensity by Province - Browse by Year" heatmap or the "Weekly Dengue Intensity by District - Browse by Year" heatmap, the chart view can remain zoomed in or shifted from its original framing. To properly view the full heatmap again, click the "Reset axes" control in the chart's toolbar, indicated by the house/home icon, which restores the chart to its default, fully zoomed out view.

8. Outbreak Analysis Section - Charts and Interpretation

This page contains four charts focused specifically on detecting and characterising outbreak or anomalous behaviour in the weekly case series.

8.1 Epidemic Channel: Weekly Cases vs. Threshold (Mean + 2 SD) with 52-Week Rolling Average

Description: The observed weekly national case count (blue line) is plotted against an "epidemic channel": a grey shaded band representing the historically expected range for each week (based on the mean and standard deviation of cases in that week across all years) and a red dotted threshold line set at the mean plus two standard deviations. Any week where the observed line rises above this threshold is filled in red as an "epidemic period." A green dashed line shows the 52 week rolling average and text annotations flag the 2017 and 2019 outbreak periods. Range selector buttons and a range slider allow zooming to 2, 5 or 10 year windows.

Interpretation: This is a classic epidemic-channel (or "epidemic corridor") chart used in public health surveillance. Weeks where the blue line breaks out of the grey expected band and crosses the red threshold are statistically unusual, they represent a level of transmission that would not be expected under normal seasonal variation and are flagged in red as epidemic periods. The chart shows that such

breakouts are episodic rather than continuous, with the 2017 and 2019 periods standing out as the most sustained and severe departures from the expected channel, while the 52 week rolling average line shows whether the underlying baseline transmission level has itself been rising or falling over the longer term, independent of any single week's spike.

8.2 Anomaly Detection — STL Decomposition + IQR Flagging (anomalize)

Description: Using the STL (Seasonal Trend decomposition using Loess) method followed by an interquartile range (IQR) based anomaly rule, this chart plots the observed weekly case series (grey line) with a shaded "expected band" derived from the decomposition. Individual weeks are marked as small blue dots ("normal week") or larger red dots with a white outline ("anomalous week") depending on whether that week's residual, after removing trend and seasonality, falls outside the expected IQR based bounds.

Interpretation: Unlike the simple threshold view in 8.1, this method statistically removes the normal trend and seasonal cycle first and only then checks whether what remains is unusually large, so it can detect an outbreak even in a week that would not look extreme in raw numbers, as long as it is unusual after accounting for what would normally be expected at that point in the trend and season. Clusters of red anomaly points concentrated around 2017 and 2019 corroborate the epidemic-channel findings using an independent statistical method, while any anomalous points appearing outside those two well known epidemics point to smaller, less obvious localised or short lived surges that a simple raw threshold view might not flag as clearly.

8.3 Cumulative Cases by Year

Description: For each year, weekly cases are accumulated (summed cumulatively week by week) and plotted as a running total against epidemiological week. Five notable years (2017, 2019, 2023, 2024, 2025) are drawn as distinct coloured lines with an end of line label, while all other years are summarised as a grey shaded interquartile range (IQR) ribbon with a dotted median line, giving a "typical year" baseline for comparison. A vertical dotted reference line marks the mid year point (week 26).

Interpretation: A highlighted year's cumulative line rising steeply above the grey "typical range" band early in the year indicates an unusually fast building epidemic, while a line that stays within or below the band for most of the year but steepens later indicates a late season surge. Comparing the final height of each highlighted line at week 52 ranks those years by total annual burden and comparing where each line diverges from the median tells the viewer not just how bad a year was overall, but when during the year the excess cases actually accumulated.

8.4 5-Year Rolling Average vs. Annual Cases

Description: A bar chart of total annual cases, with each bar coloured red if that year's total is at or above the trailing 5-year rolling average or blue if it is below. A dark line traces the 5-year rolling average itself across years, a horizontal dotted reference line marks the most recent rolling average value and an annotation highlights the single highest case year.

Interpretation: This chart separates short term, single year spikes from the underlying multi year trend. A year coloured red is running hotter than its recent five year history, even if it is not an all time

record, which can be an early indicator that the disease burden is structurally increasing rather than just experiencing one bad year. A lengthening run of red bars or a rolling average line that keeps climbing over successive years, suggests that the baseline (endemic) level of dengue transmission in Sri Lanka is itself rising over time, rather than the disease simply having occasional unrelated outbreak years against an otherwise flat background.

9. Data Explorer Section - Filter Facilities

The Data Explorer page presents the full district level weekly dataset (date, epidemiological week, district and case count) as a searchable, sortable table, together with a set of filter controls placed directly above it.

9.1 Date range filter

Two calendar picker fields, "Start date" and "End date," let the viewer restrict the table to only the rows whose date falls within the chosen range. Clicking either field opens a pop up calendar (powered by the flatpickr date-picker library) rather than requiring the date to be typed manually. Only rows with a date on or after the start date and on or before the end date, remain visible; leaving either field empty removes that side of the restriction.

9.2 Case-count range filter

Two numeric fields, "Cases" Min and Max, let the viewer restrict the table to rows whose reported case count falls within a chosen numeric range (for example, only showing weeks with 100 or more cases in a district). The filter updates when a value is entered and confirmed with Enter or when the field loses focus.

9.3 Refresh button

The "🔄 Refresh" button clears the start date, end date and both case count fields in one click, as well as any text typed into the table's own search boxes, and redraws the table with the complete, unfiltered dataset. This gives the viewer a quick way to reset all filters at once rather than clearing each control individually.

9.4 Built in table search and sorting

In addition to the custom date and case count controls above the table, the table itself (a DataTables-powered interface) provides per column search boxes in its header row for the Week and District columns, letting the viewer type a value to instantly narrow the table to matching rows. Clicking any column header sorts the table by that column and clicking again reverses the sort order.

9.5 Pagination, scrolling and export

The table displays a configurable number of rows per page (15, 25, 50 or 100), with page navigation controls beneath it. The table also scrolls horizontally on narrow screens so that no column is cut off. Two export buttons, "CSV" and "Excel," let the viewer download the currently filtered and displayed data in either format for offline analysis.

10. Summary

Taken together, the dashboard's five pages move the viewer from the broadest possible picture (national totals and long run trend on the Overview page) down to increasingly specific lenses: the recurring seasonal cycle (Seasonality & Trends), the geographic distribution of the burden (District & Province Analysis), the statistical identification of outbreak and anomalous weeks (Outbreak Analysis) and finally direct access to the underlying district level records for independent analysis (Data Explorer). Read in sequence, the charts support a consistent epidemiological narrative for Sri Lanka: dengue is an endemic, strongly seasonal disease concentrated in the populous wet zone districts, punctuated by occasional, statistically distinct epidemic years, most notably 2017 and 2019 layered on top of what the 5-year rolling average and cumulative case comparisons suggest may be a gradually rising underlying baseline.